

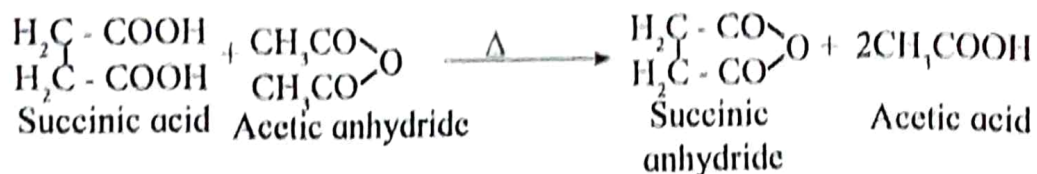
Activity No.2

Date: / /

Aim: To prepare succinic anhydride.

Theory: Succinic acid is a dicarboxylic acid which on heating loses a water molecule and undergo cyclisation reaction to form succinic anhydride.

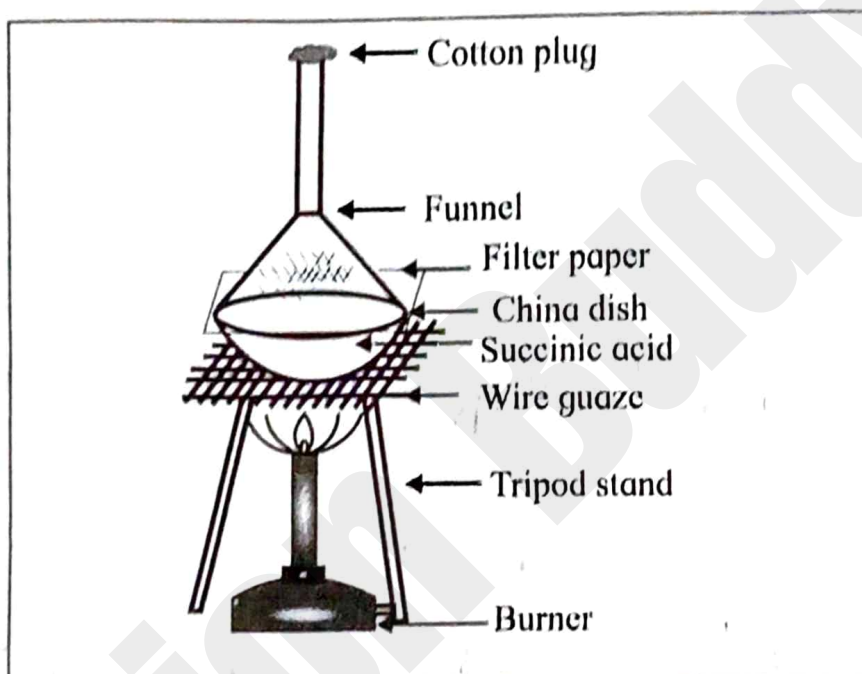
Chemicals Reaction:



Apparatus: Evaporating dish, funnel, cotton plug, tripod stand, wire gauze, pair of tongs, filter paper, etc.

Chemicals: Succinic acid.

Diagram:



Procedure:

1. Take 2 g of succinic acid in a dry evaporating dish, add 1 ml of acetic anhydride.
2. Cover the dish with filter paper having number of small holes in the centre.
3. Keep an inverted funnel on the filter paper close nozzle of the funnel with cotton plug. Place evaporating dish on the tripod stand with wire gauze as shown in above diagram.
4. Now heat the china dish slowly and carefully on low flame.
5. After sometime the acid sublimes and anhydride collects on the inner side of funnel.
6. Stop heating and remove the funnel carefully with pair of tongs and cool
7. Collect the niddle shape crystals of succinic anhydride on paper and weigh it.

Result:

1. Colour of the crystals = white
2. Shape of the crystals = Needle
3. Yield of succinic anhydride = 1.96 g

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

- The molecular formula of succinic acid is ----
 a. $C_4H_6O_4$ b. $C_6H_4O_2$
 c. $C_4H_4O_2$ d. $C_6H_6O_2$
- The by product is formed in the formation of succinic anhydride is ----
 a. water b. acetic acid
 c. acetic anhydride d. succinic acid
- The process carried out to convert succinic acid to succinic anhydride is
 a. oxidation b. reduction
 c. hydration d. dehydration
- The role of acetic anhydride in chemical reaction is as
 a. reducing agent b. oxidising agent
 c. dehydrating agent d. hydrating agent
- IUPAC name of oxalic acid is
 a. ethanedioic acid b. butanedioic acid
 c. propanoic acid d. propanedioic acid

Short answer questions

1. What happens when succinic acid undergoes dehydration?

Ans. When succinic acid undergoes dehydration it gives succinic anhydride.

2. Write IUPAC name of succinic acid.

Ans. IUPAC name of succinic acid is butanedioic acid.

3. Write names and formulae of any two dicarboxylic acid other than phthalic acid and succinic acid.

Ans. 1) Oxalic acid - $COOH$
 2) Malonic acid CH_2-COOH
 3) Adipic acid $COOH-(CH_2)_4-COOH$

4. Define the term fatty acid.

Ans. Fatty acids are monocarboxylic acids having long hydrocarbon chain especially occurring as ester in fats and oils.

5. Write chemical equation for the preparation of propionic anhydride from propionic acid.

Ans. $CH_3-CH_2-COOH \xrightarrow{P_2O_5} CH_3-CH_2-CO-O-CH_2-CH_2-COOH + H_2O$
 Propionic acid Phenyl carboxylic acid

Remark and sign of teacher: